# Python Activity-Based Learning

## Case Study 26: Nested Loop Examination Seating Generator

### 1. Problem Statement

Generate a classroom seating arrangement using row and column identifiers. Mark selected seats as reserved and display the arrangement within the room's fixed capacity.

### 2. Objectives

- Use nested loops to create a seating grid.
- Represent the arrangement with a nested list.
- Use conditions to mark reserved seats and print a simple seating chart.

### 3. Assumptions

- Rows and columns must both be positive integers.
- Seat labels use row letters and column numbers, such as A1 or B3.
- Reserved seats are supplied as comma-separated labels, for example A2,B1.

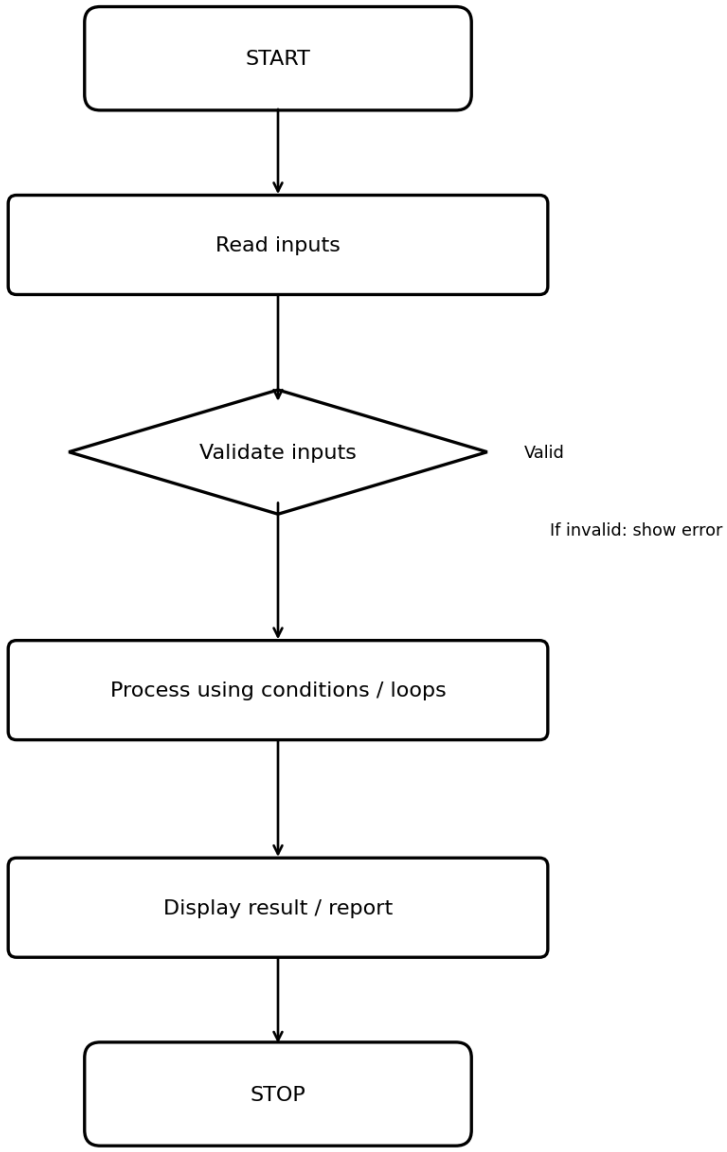
### 4. Algorithm

1. Read the number of rows and columns.
2. Validate dimensions.
3. Read reserved seat labels and store them in a list.
4. Use nested loops to build each row of the seating matrix.
5. Mark each seat as RESERVED or available.
6. Print the seating chart and capacity summary.

### 5. Flowchart

The diagram below shows the overall program flow. The validation decision represents the input checks described in the algorithm.

**Flowchart: Nested Loop Examination Seating Generator**




## 6. Pseudocode

```
START
READ rows, columns
IF rows <= 0 OR columns <= 0 THEN DISPLAY invalid dimensions
ELSE
    READ reserved seat labels
    CREATE empty nested list seating
    FOR each row index
        CREATE empty row list
        FOR each column index
            label = row letter + column number
            IF label is in reserved list THEN APPEND [label, RESERVED]
            ELSE APPEND [label, AVAILABLE]
        END FOR
        APPEND row to seating
```

```
    END FOR
    DISPLAY seating rows
    DISPLAY total capacity and reserved count
END IF
STOP
```

## 7. Python Source Code

The solution uses only basic Python features from the assigned syllabus.

```
  rows = int(input("Number of rows: "))
  columns = int(input("Seats per row: "))

  if rows <= 0 or columns <= 0 or rows > 26:
      print("Invalid dimensions. Use 1 to 26 rows and positive columns.")
  else:
      reserved_text = input("Reserved seats (comma-separated, e.g. A2,B1): ")
      reserved = reserved_text.upper().replace(" ", "").split(",")
      if reserved == [""]:
          reserved = []

      seating = []
      reserved_count = 0

      for r in range(rows):
          row = []
          row_letter = chr(65 + r)
          for c in range(1, columns + 1):
              label = row_letter + str(c)
              if label in reserved:
                  row.append(label + "[R]")
                  reserved_count += 1
              else:
                  row.append(label + "[ ]")
          seating.append(row)

      print("\nSEATING CHART ([R] = Reserved, [ ] = Available)")
      for row in seating:
          print("  ".join(row))

      print(f"Total capacity: {rows * columns}")
      print(f"Reserved seats in chart: {reserved_count}")
      print(f"Available seats: {rows * columns - reserved_count}")
```

## 8. Test Cases

| Test input / condition | Expected result | Purpose |
|---|---|---|
| 2 rows, 3 columns, A2,B1 | Six seats total | A2 and B1 marked reserved |
| 0 rows | Invalid dimensions | Error message |
| 2 rows, 2 columns, no reserved seats | Four available seats | All positions available |

## 9. Sample Output Screenshot

Sample console output for one test run:

```
C:\Users\Badri\PycharmProjects\PythonProject-1\.venv\Scripts\python.exe "C:\Users\Badri\PycharmProjects\PythonProject-1\Case Studies\case
 study-9.py"
Number of rows: 2
Seats per row: 7
Reserved seats (comma-separated, e.g. A2,B1): A1,A2,A3,A4,A5,A6,A7

SEATING CHART ([R] = Reserved, [ ] = Available)
A1[R]  A2[R]  A3[R]  A4[R]  A5[R]  A6[R]  A7[R]
B1[ ]  B2[ ]  B3[ ]  B4[ ]  B5[ ]  B6[ ]  B7[ ]
Total capacity: 14
Reserved seats in chart: 7
Available seats: 7

Process finished with exit code 0
```

## 10. Conclusion

Nested loops and nested lists make it straightforward to generate and display a classroom seating plan.

Note: Assumed rates, thresholds, and sample values are for this educational exercise and can be changed if the instructor supplies different rules.